Vivek Kumar

+1 (763) 900-2756 | Minneapolis, MN | Kuma0355@umn.edu | LinkedIn | GitHub | Portfolio

EDUCATION

University of Minnesota Twin Cities — Minneapolis, MN
Master of Science in Robotics — *Sep 2024 – Present*

National Institute of Technology Rourkela — Odisha, India
Bachelor of Technology in Mechanical Engineering — *2019 – 2023*

SKILLS

Robot Learning: Imitation Learning, VLA Policies, ACT, Diffusion Policies, Reinforcement Learning, Teleoperation
Generative & Foundation Models: Vision-Language Models, World Models, Diffusion/Flow Matching, LoRA, DPO
Programming & ML: Python, C++, PyTorch, Hugging Face, OpenCV, TensorRT, ONNX, Docker
Robotics & Simulation: ROS 2, Mujoco, Isaac Sim, ManiSkill, Linux

EXPERIENCE

Factory Intelligence | Robot Learning Intern — Lafayette, IN
Robot Foundation Models, VLAs, RL, Real-world Deployment — *Aug 2026 – Present*

- Deployed a robotic **assembly workcell** with two YAM arms for an electrical fabrication client, chaining task-specific robot policies via a learned completion flag, with high success rates and **multi-hour production** runs.
- Working on **mid-training** $\pi_{0.5}$ on **100+ h** of multi-task YAM data to improve downstream fine-tuning, applying **human-in-the-loop RL** for precise, contact-rich tasks such as **screwing** and **wire hooking**.
- Fine-tuning π-family with a **joint-torque** history token and other robot policies on **800+** episodes for **contact-rich** tasks on YAM arms and the Almond Axol, a wheeled **semi-humanoid**.
- Contributed to **FI-OS**, the company's in-house robot-learning platform and UI that runs the full pipeline: onboarding new robots, teleoperation, data collection, training, evals and deployment.

NextGX Lab, University of Minnesota | Graduate Research Assistant — Minneapolis, MN
Autonomous Driving, VLA, VLMs — *June 2025 – July 2026*

- Built a VLM model for proactive escalation in autonomous driving using Qwen3-VL and **NVIDIA Cosmos** as teacher model, fine-tuned on a **nuScenes** dataset via **LoRA** and **DPO**, achieving **92.1%** accuracy.
- Contributed to developing a sign-guided **Vision-Language-Action** framework for long-horizon **mapless navigation** that leverages semantic scene understanding and language reasoning to navigate toward user-specified destinations.

GLOVEX Lab, University of Minnesota | Graduate Researcher — Minneapolis, MN
Inverse Problems, Diffusion and Flow-matching Models — *Apr 2025 – July 2026*

- Conducted research on inverse problems using generative priors like **diffusion** and **flow-matching** models.
- Built ZS-PDP, a novel **zero-shot** blind inpainting framework, outperforming both **SOTA** supervised and zero-shot methods.

PUBLICATIONS

- **Vivek Kumar**, et al. *VLM-Driven Proactive Escalation for Teleoperations in Autonomous Vehicles.* **CVPR Precognition Workshop 2026**. **Paper**
- W Zhang, **Vivek**, et al. *Zero-Shot Blind Inpainting with Pretrained Diffusion Priors.* **ICLR 2027 (Under Review)**

PROJECTS

Multimodal Diffusion Policy for Autonomous Driving — *Python, PyTorch*

- Trained a lightweight Vision-Language-Action **(VLA) diffusion policy** on a custom driving dataset, using a CNN-Transformer architecture to predict autonomous trajectories from natural language instructions.

Robot Manipulation via Action Chunking Transformers in ManiSkill — *Python, PyTorch, ManiSkill*

- Built an end-to-end **imitation learning** pipeline using Action Chunking Transformer (ACT) architecture to train a robot-arm manipulation policy, achieving **95%+** success rate on the PickCube task with **30k** training iterations.

Real-Time Video Prediction for Low-Latency Teleoperation — *Python, PyTorch*

- Adapted and trained a **real-time video prediction** model on the **BDD100K** autonomous driving dataset to predict future frames and compensate for **100+ ms** network-induced visual latency in teleoperation.